NXTWAVE OF INNOVATION IN ADVANCE TECHNOLOGY

Department of Computer Science and Engineering • Jaipur, Rajasthan

ACADEMIC INTERNSHIP REPORT

Web Development at Codec Technologies Pvt. Ltd.



1-Month AICTE & ICAC-Approved Industrial Training & Internship Program

Internship Period: 08 July 2026 – 13 August 2026 • Mode: Pan India (Hybrid)

CANDIDATE INFORMATION

Candidate Name: Damin Sahu
Degree: B.Tech in Computer Science & Engg.
Institution: NXTWAVE of Innovation in Advance Tech
Location: Jaipur, Rajasthan
Roll Number: 25TEC2SE291
Academic Year: 2025–2029

HOST ORGANIZATION & MENTORSHIP

Host Organization: Codec Technologies Pvt. Ltd.
Location: Chandivali IT Hub, Mumbai
Designation: Web Developer Intern
Industry Mentor: Dr. Anurag Shrivastava (Prog. Mgr.)
Institutional Supervisor: Aparajita Negi (HOD, NIAT)

A formal technical report submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology in Computer Science and Engineering

NXTWAVE of Innovation in Advance Technology • 2026 • Submission Date: 22/10/2026

Internship Certificate & Candidate Declaration

Official credential issued by Codec Technologies Pvt. Ltd. and student statement of integrity



Candidate Declaration

I, **Damin Sahu**, student of *B.Tech in Computer Science and Engineering* at **NXTWAVE of Innovation in Advance Technology**, Jaipur, Rajasthan, hereby declare that the work presented in this internship report entitled "*Web Development at Codec Technologies Pvt. Ltd.*" is an authentic record of my industrial training completed during the period from **08 July 2026 to 13 August 2026**. The single web project discussed in this report, namely the **Personal Portfolio Website**, was engineered by me as part of the assigned technical curriculum under the guidance of **Dr. Anurag Shrivastava**, Program Manager, Codec Technologies Pvt. Ltd.

I further affirm that this report adheres to academic integrity standards and accurately reflects the verified technologies, codebase structure, and learning outcomes without unauthorized fabrication.

Date: 22/10/2026

Place: Jaipur, Rajasthan

Damin Sahu
Signature of the Student
Roll No.: 25TEC2SE291

Executive Abstract & Acknowledgements

Overview of industrial internship achievements and formal academic recognitions

Executive Abstract

This report documents the one-month industrial training and internship completed by **Damin Sahu** (Roll No: **25TEC2SE291**) at **Codec Technologies Pvt. Ltd.** in the capacity of **Web Developer Intern**, spanning from **08 July 2026 to 13 August 2026**. Conducted in a hybrid work environment across Pan India under the *AICTE & ICAC-Approved Industrial Training and Internship Program*, the primary objective of this internship was to bridge theoretical computer science education with contemporary front-end web engineering practices.

The core technical deliverable discussed in this report is the design, implementation, and deployment of a responsive **Personal Portfolio Website**. The application was constructed using modern web engineering principles, leveraging semantic HTML5 for structure, modular CSS3 architecture with custom design tokens for responsive aesthetics, and TypeScript/JavaScript with React and Vite for component-driven UI architecture. Source code management and collaborative versioning were conducted through Git and GitHub (**daminsahu697-debug/portfolio**), culminating in a live public deployment on Vercel (**damin-portfolio.vercel.app**).

Key learning outcomes encompass modular UI design, interactive state and modal management, asset optimization for fast page loads, cross-device responsiveness, and continuous deployment workflows. The experience provided crucial insights into software development hygiene, industry-standard toolchains, and real-world problem-solving methodologies.

Acknowledgements

I express my profound gratitude to **Codec Technologies Pvt. Ltd.** for providing an intensive, structured, and immersive industrial internship platform. I am particularly indebted to **Dr. Anurag Shrivastava**, Program Manager and Talent Acquisition Manager, for his professional leadership, program organization, and constructive feedback throughout the internship tenure.

I extend sincere thanks to **Aparajita Negi** (Head of Department, NIAT) and the academic faculty at **NXTWAVE of Innovation in Advance Technology**, Jaipur, Rajasthan, for establishing foundational principles in computer science and software development that enabled me to execute industrial tasks with technical rigor.

Finally, I appreciate the collaborative spirit of fellow peers in the July 2026 cohort whose technical exchanges and shared problem-solving discussions enriched this industrial learning journey.

> **Verification Note:** This internship was conducted under AICTE Corporate ID: `CORPORATE6759d549ce59e1733940553` and National Career Service (NCS) ID: `E19E86-0116588288923`. All document dates (08/07/2026 to 13/08/2026) are preserved exactly as stated in official records.

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Chapter 1: Introduction to Codec Technologies

Organizational background, operational scope, and quality accreditations

1.1 Organizational Profile

Codec Technologies Pvt. Ltd. is an established IT and business consultancy organization headquartered at **Chandivali IT Hub, Mumbai, India**, with an operational footprint spanning across 27+ countries. As stated in official institutional correspondence, Codec Technologies operates as a global platform committed to empowering emerging technical learners and connecting talent with industry opportunities through structured training programs. The organization trades online via its official portal (www.codectechnologies.in).

1.2 Corporate Credentials & Accreditations

The host organization holds verified industrial recognitions that reinforce the legitimacy and technical rigor of its training curricula:

- **Quality Certification:** ISO 9001:2015 certified organization, ensuring standardized operational quality.
- **Industry Partnerships:** Recognized Google for Education Partner, aligning training with industry-standard development ecosystems.
- **AICTE & ICAC Approval:** Operating under AICTE Corporate ID: `CORPORATE6759d549ce59e1733940553`.
- **National Career Service Registration:** Verified NCS ID: `E19E86-0116588288923`.

1.3 Mentorship & Evaluation Framework

The industrial internship program is administered under the leadership of **Dr. Anurag Shrivastava**, Program Manager and Talent Acquisition Head. The cohort framework incorporates milestone reviews, structured engineering tasks, and evaluation criteria designed to simulate agile development practices.

Chapter 2: Internship Work Profile & Methodology

Internship objectives, role definition, and professional web engineering methodology

2.1 Internship Objectives

The primary purpose of the internship at Codec Technologies was to provide structured industrial exposure to contemporary web development paradigms. Key objectives included:

- Translating academic computer science foundations (algorithms, object-oriented concepts, database structures) into functional client-side web software.
- Developing a production-ready web application adhering to semantic conventions, responsive UI/UX principles, and accessibility standards.
- Mastering version control workflows using Git and GitHub for code traceability and team collaboration.
- Understanding continuous deployment, static hosting mechanics, and production build pipelines.

2.2 Role Definition & Operating Mode

The candidate served in the formal capacity of **Web Developer Intern** within the July 2026 cohort (duration: **08 July 2026 to 13 August 2026**). Operating in a **Pan India, Hybrid Mode**, tasks were completed through remote development cycles, online repository submissions, and virtual technical coordination.

2.3 Professional Engineering Lifecycle

The technical workflow followed during the internship reflected established industry practices, structured into five progressive phases:

Workflow Phase	Key Engineering Activities & Deliverables
1. Requirement Analysis	Deconstructing technical requirements, user stories, content hierarchies, and asset catalogs.
2. Interface Architecture	Designing modular layout skeletons, typography systems, color tokens, and responsive breakpoints.
3. Component Development	Implementing modular React/TypeScript components, interactive state hooks, and DOM event listeners.
4. Responsive Testing	Validating cross-device viewport behavior, checking touch interactions, and auditing console logs.
5. Deployment & Release	Configuring build optimizations in Vite, committing clean git trees, and publishing to Vercel hosting.

2.4 Source Code Management Standards

A critical component of the industrial workflow was rigorous version control. Code changes were tracked using atomic Git commits with clear imperative messages, branch management, and remote repository synchronization on GitHub. Sensitive configuration files such as environment credentials (`.env`) and build caches were systematically excluded using a comprehensive `.gitignore` configuration.

Chapter 2 (Cont.): Personal Portfolio Website Project

Comprehensive technical overview of the primary industrial project deliverable

2.5 Project Overview & Purpose

The primary software deliverable completed during the internship was the **Personal Portfolio Website**. The objective was to design and deploy a personal web platform that professionally showcases the developer's academic profile, technical skill inventory, featured projects, extracurricular achievements, and verified contact channels.

> **Verified Project Repository & Live Deployment:**
> • **GitHub Repository:** https://github.com/daminsahu697-debug/portfolio
> • **Live Production URL:** https://damin-portfolio.vercel.app
> • **Developer Role:** Frontend Developer • **Build Status:** Production Verified (0 Errors)

2.6 Component Hierarchy & Modular Structure

The website is engineered as a component-driven Single Page Application (SPA) following a strict separation of concerns:

- **Application Shell (`App.tsx`):** Serves as the central view orchestration layer, hosting global navigation, theme contexts, main content sections, and the footer.
- **Navigation & Accessibility (`Navbar.tsx`):** Provides smooth anchor-based scrolling across sections, responsive mobile drawer controls, and quick resume access.
- **Central Data Decoupling (`src/data/portfolio.ts`):** All resume data, academic milestones, project metadata, and contact links are isolated in a typed data model, ensuring that UI components remain pure and reusable without hardcoded text.
- **Theming & Context Engine (`ThemeContext.tsx`):** Manages dark/light mode switches alongside dynamic color accent preferences, persisting settings to local storage.

2.7 Core Functional Sections Implemented

- **Hero Introduction:** Features personal positioning ("*Building Practical Software Solutions Across Web, AI & IoT*"), embedded circular portrait, and quick CTA buttons.
- **Education Timeline:** Details B.Tech CSE at NXTWAVE of Innovation in Advance Technology (CGPA: 9.33, 2025–2029), Class XII (82.8%), and Class X (97%).
- **Categorized Technical Skills:** 6 organized clusters covering Programming Languages, Web Technologies, AI & CV, Embedded IoT, Backend & Databases, and Core CS Concepts.
- **Verified Projects Showcase:** Displays project role, technologies, and verified repository links for *SecureChain DMS* and the *Portfolio Website*.
- **Integrated Resume Hub:** Modal viewer enabling inline PDF inspection and one-click downloading of `Damin_Sahu_Resume.pdf`.
- **Theme Customizer & Accessibility:** Persistent light/dark toggles and multi-hue accent selections respecting system reduced-motion preferences.

Chapter 3: Implementation Details & Learning Outcomes

Engineering execution, verification testing, and technical competencies acquired

3.1 Frontend Implementation Process

The implementation prioritized clean software design, maintainability, and zero unnecessary dependencies. The interface was engineered with:

- **Semantic Structure:** Developed using HTML5 semantic elements (`<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<footer>`) to guarantee accessible document outlining and optimal search indexing.
- **Modular Vanilla CSS Design System:** Constructed entirely in `index.css` utilizing CSS Custom Properties (CSS variables) for design tokens (colors, font scales, radius, shadows). This eliminated heavy CSS frameworks while granting full styling flexibility.
- **Component Architecture in React 19 & TypeScript:** Strict TypeScript interfaces were defined for all entities (`Project`, `EducationItem`, `ResumeConfig`), guaranteeing type safety and eliminating runtime null reference bugs.
- **Centralized Content Decoupling:** All website copy, credentials, project descriptions, and links are isolated in `src/data/portfolio.ts`, separating content from UI presentation.

3.2 Testing, Verification & Quality Assurance

Rigorous quality assurance was carried out before deployment to ensure cross-device consistency and performance:

Testing Dimension	Verification Methodology & Observed Outcome
Viewport Responsiveness	Tested across standard desktop (1920x1080), laptop (1366x768), tablet (768x1024), and mobile viewports (375x667). Confirmed fluid reflow without horizontal scrolling.
Static Asset Delivery	Verified HTTP 200 responses for static image assets (`profile.jpg`: 169.8 KB) and downloadable resume (`resume.pdf`: 76.5 KB) via production CDN.
Build & Compilation	Executed `tsc -b` and `vite build`. Produced production bundle of 281.9 KB JS and 39.3 KB CSS transformed across 1,950 modules with 0 errors.

3.3 Key Learning Outcomes & Professional Growth

- **Technical Competency:** Solidified mastery of modern JavaScript/TypeScript, React component design, Vite bundler optimization, and CSS layout algorithms (Flexbox, CSS Grid).
- **Software Hygiene:** Developed an appreciation for strict git hygiene, atomic commit messages, and automated deployment pipelines.
- **Problem-Solving Maturity:** Gained capability in debugging layout anomalies, managing modal escape keys and focus states, and implementing seamless theme persistence using `localStorage`.

Chapter 4: Technical Challenges, Conclusion & Suggestions

Engineering hurdles resolved, academic synthesis, and recommendations for future cohorts

4.1 Engineering Challenges & Solutions

During the execution of the portfolio web project, several practical technical hurdles were encountered and systematically resolved:

- **Challenge 1: Cross-Device Responsive Layout Stability:** Ensuring that complex cards with varying text lengths, badges, and image portrait containers aligned evenly across both narrow mobile viewports and wide displays.
  Resolution: Implemented fluid CSS Grid with `minmax()` functions, combined with CSS `clamp()` for responsive typography, preventing layout breakage without excessive media query rules.
- **Challenge 2: Multi-Theme Contrast & Modal Accessibility:** Maintaining high-contrast text readability across light, dark, and colored accent themes while ensuring the PDF viewer modal was fully accessible.
  Resolution: Formulated a centralized semantic color token palette with WCAG AA compliance checks and implemented backdrop overlay click listeners and keyboard dismiss handlers.
- **Challenge 3: Production Build Asset Resolution:** Ensuring that static PDF documents and image headshots loaded seamlessly on both local development servers and public cloud hosting without path mismatches.
  Resolution: Standardized asset routing under the Vite `public/` directory and verified deployment HTTP headers using production curl requests.

4.2 Conclusion & Curriculum Alignment

The industrial training completed at **Codec Technologies Pvt. Ltd.** served as an invaluable bridge between academic theory and real-world software engineering. Concepts introduced during the B.Tech Computer Science and Engineering program at **NXTWAVE of Innovation in Advance Technology** — such as software development lifecycles, structured programming, and modular design — were directly put into practice. The resulting portfolio website stands as an active, verified testament to the front-end competencies, design sensibilities, and deployment capabilities acquired during this program.

4.3 Constructive Suggestions for the Internship Program

- **Structured Collaborative Review:** Incorporating mandatory peer pull-request reviews would expose interns to team code critique and merge conflict resolution.
- **Extended Timeline:** While the 1-month duration is intensive and focused, expanding the window to 6–8 weeks would permit deeper integration of backend microservices and database persistence.
- **Automated CI/CD Workshops:** Introducing hands-on sessions on GitHub Actions for automated linting and unit testing would further elevate industry readiness.

Annexures & Document References

Official internship offer letter, institutional verification IDs, and public repository links

Annexure I: Official Internship Offer Letter



CODEC TECHNOLOGIES

NATIONAL INTERNSHIP PORTAL

AICTE ID - CORPORATE6759d549ce59e1733940553

INTERNSHIP OFFER LETTER

Dear Damin Sahu,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Web Developer Intern
- Location & Mode: Pan India, Hybrid
- Duration : 08/07/2026 to 13/08/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.
Best regards,

Sincerely,
Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 08/07/2026

Codec Technologies ISO-9001:2015 ©All Rights are Reserved Website:codectechnologies.in

Dr. Anurag Shrivastava
Talent Acquisition Manager

Google for Education Partner

ISO 9001 CERTIFIED

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Annexure II: Official References & Verified Project Links

Reference Item	Verification Details & Official Resource URL
Host Organization	Codec Technologies Pvt. Ltd. • Portal: https://www.codectechnologies.in
National Career Service	Official NCS ID: `E19E86-0116588288923` • Ministry of Labour & Employment, Govt. of India
AICTE Corporate ID	Approved Corporate Registration ID: `CORPORATE6759d549ce59e1733940553`
GitHub Repository	Portfolio Source Code: https://github.com/daminsahu697-debug/portfolio
Live Production Website	Public Portfolio URL: https://damin-portfolio.vercel.app